

You are a proficient Java programmer and familiar with various algorithms.

Your task is to implement a Java code given a natural language description illustrating an algorithm and a Java signature.

Implement the `calculate` method in Java. It must use a stack to correctly handle nested parentheses () while also supporting + and - operations.

Please implement the function with the following template.

```
java ⌵ Collapse Copy  
1  ```java  
2  class Solution {  
3  public int calculate(String s) {  
4      return result;  
5  }  
6  }  
7  }
```

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```
1  Enter your code here...
```

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```
java ⌵ Collapse Save Copy  
  
import java.util.Stack;  
  
  
7      int number = 0;  
8      int sign = 1;  
9  
10     for (int i = 0; i < s.length(); i++) {  
11         char c = s.charAt(i);  
12         if (Character.isDigit(c)) {  
13             number = number * 10 + (c - '0');  
14         } else if (c == '+') {  
15             result += sign * number;  
16             number = 0;  
17             sign = 1;  
18         } else if (c == '-') {  
19             result += sign * number;  
20             number = 0;  
21             sign = -1;  
22         } else if (c == '(') {  
23             stack.push(result);  
24             stack.push(sign);  
25             result = 0;
```

```
6  
7     return result;  
    }  
}
```

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```
1 Enter your code here...
```

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java

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```
import java.util.Stack;
```